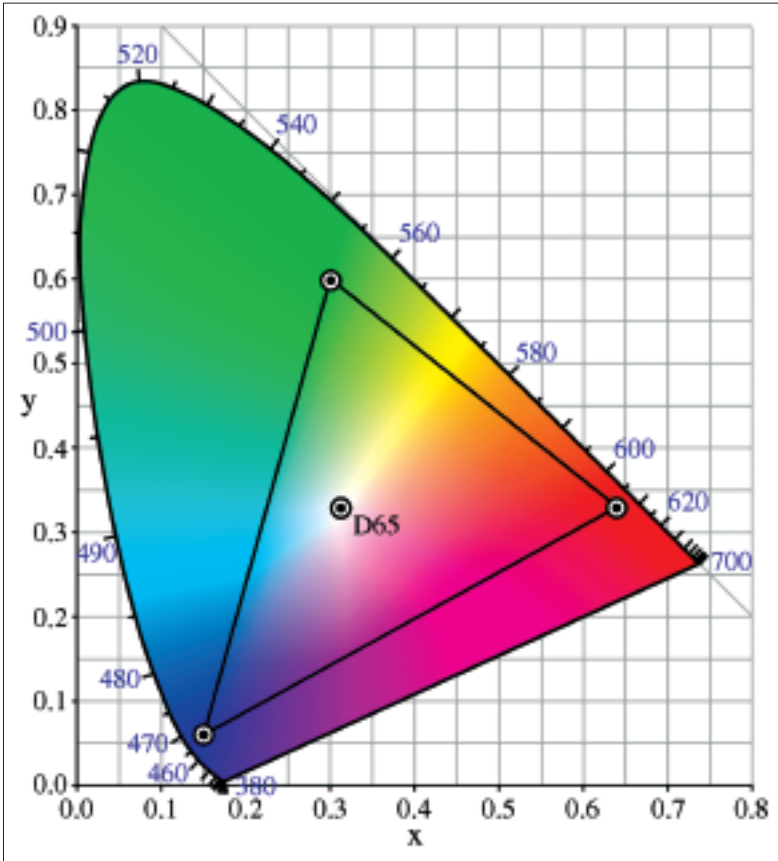


Representing Colour Using Colour Spaces

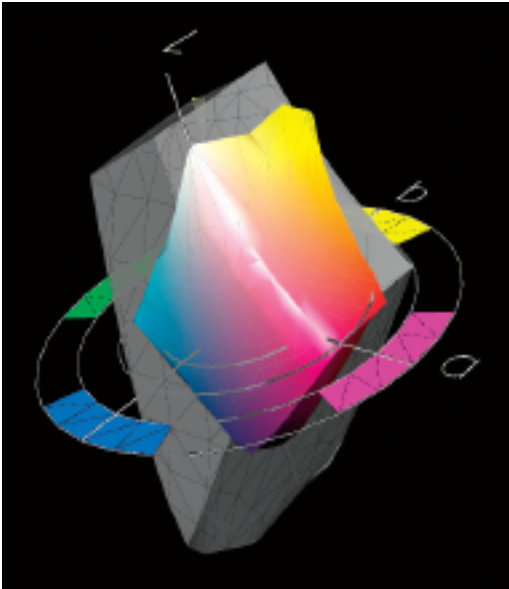
One of the most confusing things you may ever see in a text book on colour is the 2D representation of the 1931 CIE XYZ colour chart for RGB displays. It is rarely explained that this colour chart represents a plan view or projection from above the white point of a 3D volume and is shown this way purely for convenience, given the difficulty of representing 3D objects in print. All colour spaces are three dimensional volumes. The vertical axis of these spaces always represents luminance or brightness and runs from black at the bottom, through various shades of grey, to white at the top.



The most commonly used reference space today is the 1976 CIE L*a*b* (pronounced as elstar ehstar, beestar). Rather confusingly this is often referred to as Lab but there is an earlier space, the 1948 Hunter Lab space, to which that nomenclature more properly belongs. L*, standing for Luminance is the vertical axis and the orthogonal axes a* and b* are red-to-green and blue-to-yellow. L*a*b* is often used as a reference colour space for performing transformations from one colour space to another because, at present, it is the space that comes close to perceptual linearity. For example the popular Adobe applications such as Adobe Photoshop use a version of L*a*b* as their internal colour model.

The CIE XYZ 2D projection. The D65 whitepoint is marked and the triangle shows the extent of the sRGB gamut. Light wavelength is marked around the edge of the coloured area which represents the gamut of average human vision. The Z or luminance axis is at right angles to the paper

Here's a 3D plot of two superimposed gamuts in the L*a*b* colour space



Saturation, Hues, Tints And Shades

A special vocabulary is used for describing colour. Unfortunately these terms are often misunderstood or misused and in some cases are hard to define. They are saturation, hue, shade and tint.

The strongest, purest colour is said to be saturated. For displayed digital images fully saturated colours are represented by the largest numbers, or 255 for a 24-bit image (32-bit colour usually uses the extra 8 bits to represent other image attributes such as transparency). Very pure saturated colours are quite rare in nature because most naturally occurring colours are mixtures of more than a single wavelength. Artificially produced, single wavelength colours, are fully saturated.

Colour hue is the attribute that describes what appears to be the colours dominant wavelength. For example the hue for pink would be red.

For mixing opaque pigments, as in paint, tints are achieved by mixing a colour with white, while shades are achieved by mixing black with a colour. For example light pink is red mixed with white, while dark pink is black mixed with red. For colour produced by emitted light, such as from a display screen it is always a question of mixing the colour with white because the base state of all displays is black. For displays, the shade attribute of a colour can be regarded as brightness or luminance.

Colour Gamuts

Every colour device has a gamut, a range of colours it can capture or reproduce. This gamut can be represented as a device dependent colour space within the framework of any

of the reference colour spaces. The term gamut is used to refer to the relative volume of the colour space. It is often also used to refer to the area of the 2D projection of a colour space, since this is related to the volume and represents the range of saturated colours. The human visual system - for most people - has a large gamut. Frequently this is referred to as containing 65 million colours. The human visual system is not good at recognising absolute colours, but is much better at recognising the difference between discrete patches of similar shades. The figure of 65 million recognisable shades has been determined by experiment as the average colour discrimination for most people.

Colour Intent

Colour intents are one of the most difficult aspects of colour management to understand. Users of colour management systems are required to choose which intent to use. Intents are needed because none of the current elements in a colour capture, manipulation and reproduction chain can match the gamut of the human visual system. All capture devices, displays and printers have a smaller gamut. Colour printers usually have the smallest gamut, or device colour space, of all the colour devices we use. Some of the colours we can see cannot be captured, and some of the colours that a camera or scanner can capture, cannot be reproduced. This leaves us with the problem with what to do with the colours we have captured in an image but cannot reproduce in print and here is where the choice of intent comes in.

A colour intent describes how any colours that fall outside the gamut of a reproduction device, normally a printer, are rendered. There are four intents in common use; perceptual, relative colorimetric, absolute colorimetric and saturation.

Perceptual rendering compresses the source gamut so all the out-of-gamut colours fit inside the destination gamut. This can result in some distortion of the relationships between colours. With perceptual rendering none of the original colour information is lost and in theory the transformation could be reversed in order to restore the original image.

Relative colorimetric rendering maintains a near exact relationship between in-gamut colours but may clip colours that are out of gamut. Clipped colours are lost and cannot be recovered.

Perceptual or relative colorimetric intents are the best choices for photo realistic images.

The absolute colorimetric intent is similar to relative, in that it clips out-of-gamut colours and preserves those that are in gamut, but it treats the white point differently. With

absolute the white point does not change, while with relative it may, if this is required to maintain the relationship between colours. This means that with a relative conversion the appearance of whites in an image may change, they may get slightly more red or blue, while with absolute they won't. Absolute rendering is used when it is important that the colours that are reproduced remain as accurate as possible.

Saturation rendering tries to preserve the saturation of colours while making no attempt to reproduce photo realistic images. It is a good choice for business graphics such as charts.

The Colour Management Workflow

A colour management system consists of a number of hardware devices, each with their own gamut or colour space, with an image in the form of digital data being passed from one device to another. As the data moves between each device it often needs to be translated from one colour space to another. To do this it is necessary to know the source and destination colour spaces used, the devices must be calibrated and a profile that describes the devices' colour gamut must be used.

A typical colour system chain consists of a camera or scanner passing data to a computer running a colour management application. Edited colour images from the colour management application are output to a web page, a local colour printer or sent to a commercial colour press. A calibrated and profiled colour display is used to view the colour images at various stages in the process.

A properly calibrated and profiled colour display is absolutely vital to establishing colour control because it is the user's window on the entire process. These must be calibrated and profiled using a hardware monitor calibrator such as the ColorVision Spyder 2. Older displays that have been in use for several years are often unsuitable because they can no longer generate the peak brightness that is required for accurate calibration. Although LCD monitors are now the usual display type many of them are unsuitable for serious colour work, either because they lack the necessary user controls for calibration, or are limited to only 18-bit colour rather than 24-bit.

Colour And Computers

The majority of computer users expect to be able to use a digital camera or a scanner to capture an external scene, to edit and manipulate those images and then to print or display them on-screen without any unexpected changes in colour balance. Most people would probably say that they would expect a reproduced image on screen or printed on paper to "look the same" as the original scene. It would be very bad for business if, when operated by

